

U4370 HW3

Code ▼

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```
data(mtcars)
head(mtcars,2)
```

	mpg <dbl>	cyl <dbl>	disp <dbl>	hp <dbl>	drat <dbl>	wt <dbl>	qsec <dbl>	vs <dbl>	am <dbl>
Mazda RX4	21	6	160	110	3.9	2.620	16.46	0	1
Mazda RX4 Wag	21	6	160	110	3.9	2.875	17.02	0	1

2 rows | 1-10 of 11 columns

Consider Models

$$Y = \phi_0 + \phi_A A + \varepsilon$$

$$M = \alpha_0 + \alpha_A A + \nu$$

$$Y = \theta_0 + \theta_A + \theta_1 M_1 + \eta$$

1.a

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```
med.fit <- lm(hp ~ wt,data = mtcars)
out.fit <- lm(mpg ~ hp + wt, data = mtcars)
dc.fit <- lm(mpg ~ wt,data = mtcars)
# coefficients
alpha.A <- coef(med.fit)['wt']
phi.A <- coef(dc.fit)['wt']
theta.A <- coef(out.fit)['wt']
theta.1 <- coef(out.fit)['hp']
# effects
NIE.DM <- phi.A - theta.A
NIE.PM <- alpha.A * theta.1
data.frame('Difference Method' = NIE.DM, 'Product Method' = NIE.PM)
```

	Difference.Method <dbl>	Product.Method <dbl>
wt	-1.466641	-1.466641
1 row		

1.b

For every 1,000 lb increase in wt, the mpg decreases by approximately 1.467 miles per gallon specifically because the heavier weight leads to higher horsepower, which in turn consumes more fuel. This represents the portion of the weight-mpg relationship that “flows through” the engine’s power requirements.

2.a

$$H_0 : \alpha_A \theta_1 = 0 \text{ vs } H_1 : \alpha_A \theta_1 \neq 0$$

2.b

For Sobel Test

$$T_{Sobel} = \frac{\hat{\theta}_1 \hat{\alpha}_A}{\sqrt{\hat{\theta}_1^2 V(\hat{\alpha}_A) + \hat{\alpha}_A^2 V(\hat{\theta}_1)}} \sim N(0, 1)$$

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```
med.tb <- summary(med.fit)
out.tb <- summary(out.fit)
var.alpha.A <- med.tb$coefficients['wt','Std. Error']^2
var.theta.1 <- out.tb$coefficients['hp','Std. Error']^2
T.Sobel <- alpha.A * theta.1 / sqrt(alpha.A^2 * var.theta.1 + theta.1^2 * var.alpha.A)
pnorm(T.Sobel) * 2
```

```
wt
0.004554239
```

Since the p-value is less than 0.05, we reject the null hypothesis. There is strong statistical evidence that horsepower significantly mediates the relationship between vehicle weight and fuel efficiency.

For Joint Significance Test

$$T_{JST} = \frac{\hat{\alpha}_A}{V(\hat{\alpha}_A)} \wedge \frac{\hat{\theta}_1}{V(\hat{\theta}_1)} \sim N(0, 1)$$

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```
med.tb$coefficients['wt','Pr(>|t|)']
```

```
[1] 4.145827e-05
```

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```
out.tb$coefficients['hp','Pr(>|t|)']
```

```
[1] 0.001451229
```

Since both individual paths are significant (<0.05), we conclude there is a significant joint indirect effect.